Sonok Mahapatra

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Education

University of Maryland, College Park *Expected December 2027*

B.S. Computer Science & B.S. Applied Mathematics; Minor in Computational Finance *GPA: 3.7*

Relevant Coursework: Grad High Performance Computing, Grad Stochastic Processes, Grad Signal Processing, Probability Theory, Advanced Statistics, PDEs, Advanced Calculus I/II, Portfolio Management, Financial Markets, Machine Learning, Advanced Algorithms, Applied Linear Algebra, Graph Theory, Abstract Algebra, Computer Systems, Programming Languages

Awards & Certifications

Putnam Top 700 Scorer (14/120); Five-time AIME Qualifier ('18–'22); Jane Street Estimation Challenge Top 3; AMC Score (132/150); Bridgewater Prediction Market Competition (Top 35 World); Dean's List; 2nd Place Lockheed Martin CodeQuest; Presidential Scholar; Bloomberg Financial Fundamentals Certification; AWS Certified Cloud Practitioner

Experience

Amazon Web Services May 2026 – Aug 2026

Software Development Engineer Intern, Networking Infrastructure

- Building **low-latency distributed systems** and high-throughput data plane software on the AWS Networking Infrastructure team, powering AWS's global network fabric at hyperscale.
- Developing performance-critical **C/C++** networking components, optimizing kernel and userspace data paths for sub-microsecond latency and line-rate packet throughput across production network infrastructure.
- Profiling and tuning systems-level code for **cache locality**, **lock-free concurrency**, and tail-latency reduction across distributed services, applying the same performance-engineering toolkit used in market data and execution systems.

Probanker May 2025 – Aug 2025

Software Engineering Intern

- Optimized data pipelines for **interbank market simulations** — batch ingestion, feature generation, and scenario rollups — cutting end-to-end runtime by **40%** via vectorization and **C/C++ extensions**.
- Parallelized the bank simulation engine using **OpenMP** and cache-aware memory layouts to handle large concurrent scenario runs simulating interbank lending, deposit flows, and liquidity stress across 1K+ users.
- Hardened async job execution (**Celery**) and optimized ORM-heavy query paths in a multi-tenant

Western Connecticut State University Jun 2020 – Aug 2022

Undergraduate Researcher

- Published four peer-reviewed papers applying **wavelet-based signal processing** and post-quantum cryptographic methods to decentralized systems; presented at Gnoscience, Springer, ICSPIC, and AMS.
- Built multiresolution signal processing and cryptographic encoding tools in **MATLAB**; validated system resilience through simulation across noise and adversarial attack parameters.

State Street May 2023 – Aug 2023

Lead Project Developer, Equity Alpha Research

- Co-led an equity **sentiment alpha** research initiative, constructing NLP-based signals from 10K+ daily SEC filings, macroeconomic releases, and market microstructure data via scalable Google Cloud ingestion pipelines.
- Developed and backtested **BERT** and **VaderSentiment** NLP factor models, reducing out-of-sample MSE by **20%** and improving predictive information ratio over baseline.

Projects

Live Systematic Equity Trading System — Smith Investment Fund SIF Website

- Built a composite microstructure signal (IAM) from Kyle λ, Amihud illiquidity, bid–ask spreads, and inverse turnover; top quintile showed **2.3×** higher realized volatility and consistent pre-earnings spikes 3–5 days before announcements.
- Wrote a Python alpha class framework to backtest and deploy systematic equity signals live via **Alpaca**; strategies ran on a $50K paper portfolio with automated position sizing and stop-loss logic.
- Best strategy achieved a **1.8 Sharpe** over 18 months with 12% max drawdown; results held after FF5 + momentum controls, documented in a research paper submitted to fund leadership.

Technical Proficiencies

Languages: Python (NumPy, Pandas, StatsModels, SciPy), C/C++, Rust, SQL, MATLAB, R, Java, OCaml
Quant & ML: PyTorch, Scikit-learn, XGBoost, PyPortfolioOpt, TA-Lib, yfinance, Alpaca Trade API, statsmodels
Infrastructure: Docker, Kubernetes, Google Cloud Platform, PostgreSQL, Redis, Django, FastAPI, Git, GitHub Actions